Spatiotemporal calibration of wind power estimates based on wind speed reanalysis data

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January 2026

Abstract

Spatially aggregated wind power estimates derived from atmospheric reanalysis are widely used in energy system studies; however, these products often exhibit systematic biases and lack coherent uncertainty quantification across space and time. Discrepancies between reanalysis-based power estimates and observed generation provide valuable information for correcting these deficiencies. Building on this idea, we propose a spatiotemporal hierarchical Bayesian model to calibrate wind power derived from wind speed reanalysis against operational data. The proposed framework delivers spatially consistent bias corrections while jointly characterising temporal dependence and spatial correlation in the residual structure. This allows uncertainty to be propagated in a physically and statistically coherent manner across regions and time periods. Using wind power data from the United Kingdom, the approach produces calibrated power estimates with improved reliability and spatial coherence compared with standard bias-correction methods. The resulting framework provides a flexible and interpretable foundation for uncertainty-aware, large-scale wind power assessment based on reanalysis data.

Keywords. wind power; bias correction; calibration; hierarchical Bayesian model

1 Introduction

Reanalysis datasets, gridded, physically consistent reconstructions of past atmospheric states produced by assimilating historical observations into a numerical weather prediction model, have become a central tool in climate research and renewable energy assessment. The most widely referenced product at the time of writing is ERA5, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) at approximately 31 km horizontal resolution with hourly output from 1940 onwards (Hersbach et al. 2020). Earlier generations include ERA-Interim (Dee et al. 2011), MERRA-2 (Gelaro et al. 2017), and the NCEP Climate Forecast System Reanalysis (CFSR) (Saha et al. 2010).

Despite their convenience and global coverage, reanalysis products carry systematic biases relative to in-situ measurements. These biases arise from several sources: insufficient horizontal resolution to resolve local topography, simplified physical parameterisations of the atmospheric boundary layer, sparse observational coverage in the assimilation system, and representativeness mismatches between point measurements and grid-cell averages. The magnitude and spatial structure of the biases depend on the variable of interest (wind speed, solar irradiance, precipitation, temperature), the terrain type (offshore, flat onshore, coastal, mountainous), and the height above ground.

Correction methods range from simple linear or multiplicative scaling applied at the country or regional level through to fully distributional approaches such as quantile mapping (QM) and its variants, including equidistant QM (Li, Sheffield and Wood 2010), quantile delta mapping (Cannon, Sobie and Murdock 2015), and scaled distribution mapping (Switanek et al. 2017). Multivariate extensions (Cannon 2017; Vrac and Friederichs 2015; Mehrotra and Sharma 2016) address inter-variable dependence. More recently, Bayesian hierarchical frameworks have been proposed that explicitly propagate correction uncertainty (Carter, Chacón-Montalván and Leeson 2025) and data fusion models that assimilate reanalysis alongside sparse station observations (Villejo et al. 2025).

Applications of calibrated reanalysis to renewable energy span wind power simulation (Staffell and Pfenninger 2016; Olauson 2018; Benmoufok et al. 2024), solar photovoltaic (PV) estimation (Pfenninger and Staffell 2016; Wilczak et al. 2024), offshore wind resource assessment (Hayes, Stocks and Blakers 2021; Murcia et al. 2022), wave energy (Reguero et al. 2012; Pérez-Molina and Carta 2025), and hydrological modelling (Berg et al. 2003; Teutschbein and Seibert 2012). A complementary strand of work investigates the statistical properties of renewable output time series, including extreme low-wind events, seasonal variability, and long-term trends, directly relevant to power system planning (Cannon et al. 2014; Potisomporn, Adcock and Vogel 2023; Bloomfield et al. 2016).

Several cross-cutting gaps emerge from this literature. First, bias correction methods applied to the mean wind speed or irradiance distribution do not necessarily correct the temporal autocorrelation structure, which is important for quantifying periods of sustained low generation (Davidson and Millstein 2022). Second, most validation studies focus on onshore or offshore flat terrain; performance in complex mountainous terrain remains a distinct challenge (Gualtieri 2022; Contreras et al. 2026). Third, treatment of extremes requires specific distributional choices, including modifications to the tail behaviour of the corrected distribution (Turkman et al. 2021; Lee et al. 2025; Byun and Hamlet 2024). Fourth, the spatial resolution at which correction factors are estimated matters: country-level corrections can mask large within-country heterogeneity (Wang et al. 2026; Benmoufok et al. 2024). Finally, a gap persists between the statistical tools available for spatial modelling, including SPDE-based Gaussian random fields (Lindgren, Rue and Lindström 2011a; Ingebrigtsen, Lindgren and Steinsland 2014; Villejo et al. 2025), and their routine adoption in operational reanalysis calibration.

2 Background

2.1 What is a reanalysis?

A reanalysis is the output of a data assimilation system in which a fixed numerical weather prediction (NWP) model is run over a historical period, ingesting all available observations at each time step to constrain the model state. The result is a spatially complete, temporally consistent gridded archive of atmospheric variables. Because the model and assimilation scheme are held constant over time, a reanalysis is free from discontinuities introduced by changes to operational forecast systems, making it suitable for climate trend analysis.

ERA5 is the current reference product for most energy applications (Hersbach et al. 2020). Its principal improvements over ERA-Interim include a finer grid (≈31 km versus 80 km), hourly output, an ensemble of 10 members at half resolution for uncertainty estimation, and a decade of model development. MERRA-2, produced by NASA, provides an independent global reanalysis at comparable resolution (Gelaro et al. 2017). CFSR and its successor CFSv2, from NCEP, are also used, particularly in hydrological applications (Saha et al. 2010; Fuka et al. 2014).

2.2 Sources and structure of reanalysis bias

Reanalysis biases in near-surface wind speed are well documented. A systematic review across 15 global and regional products evaluated at 322 locations found that ERA5 is sufficiently reliable over offshore and flat onshore sites but carries large errors at coastal and mountainous locations (Gualtieri 2022). Studies using tall masts confirm that ERA5 can achieve correlation coefficients above 0.95 and biases within 1% at flat offshore sites (e.g. the FINO3 platform in the North Sea), while mountain sites exhibit large underestimations attributable to the coarse resolution (Gualtieri 2021). Brune, Keller and Wahl (2021) compared ERA5, MERRA-2, and the regional reanalysis COSMO-REA6 at 14 Central European sites and found that MERRA-2 systematically overestimates wind speed over land, whereas ERA5 and COSMO-REA6 are more comparable. A multi-reanalysis comparison across 77 instrumented tall towers concluded that ERA5 offers the best daily-scale correlation but that no product stands out for seasonal mean winds (Ramon et al. 2019).

For solar irradiance (global horizontal irradiance, GHI), ERA5 achieves a mean bias of approximately $+4\,\mathrm{W\,m^{-2}}$ globally , a reduction of 50,75% relative to ERA-Interim and MERRA-2 , but degrades at coastal sites and mountains (Urraca et al. 2018). Validation of satellite-derived and model-derived irradiance shares a fundamental complication: the spatial mismatch between a point observation and a grid-cell average introduces apparent errors that are not genuine model errors, and this effect is particularly pronounced at short time scales and for direct normal irradiance (Qin et al. 2022).

Biases in diurnal wind cycles are a structural weakness of coarse-resolution products. Davidson and Millstein (2022) demonstrated, using seven years of hourly plant-level wind generation records across Texas, that ERA5 and MERRA-2 show systematically declining accuracy through the evening transition period, with a marked improvement after sunrise. This diurnal pattern, also present in the high-resolution HRRR product, is attributed to misrepresentation of nocturnal boundary layer conditions. Holtslag et al. (2013) document the theoretical difficulty of parametrising the stable atmospheric boundary layer in NWP models, providing physical context for these operational errors.

2.3 Statistical methods for bias correction

A substantial methodological literature has developed around the correction of climate models and reanalysis outputs. The principal families of method are summarised below.

2.3.1 Simple scaling.

The simplest approach applies a multiplicative or additive correction derived from the long-term mean bias at each location. Staffell and Pfenninger (2016) derived national correction factors for MERRA and MERRA-2 wind output across 23 European countries; after calibration, national hourly wind output could be modelled with $R^2 > 0.95$. Benmoufok et al. (2024) systematically investigated the spatial and temporal granularity at which such correction factors should be estimated, finding that cluster-based spatial corrections across 4,834 Danish turbines reduced simulation error by up to 43%, with bimonthly temporal updating providing a further 3% improvement.

2.3.2 Linear regression and quantile mapping.

Linear regression (LR) and quantile mapping (QM) are the two most widely applied correction methods in the literature. QM aligns the empirical cumulative distribution function (CDF) of the model output with that of the observations by constructing a monotone transfer function. Campos et al. (2022) compared LR and QM for ERA5 wind speeds in the Atlantic Ocean and found that

LR performs better for low-to-moderate wind intensities while QM reduces root mean square error (RMSE) by approximately 10% and bias by approximately 50% for winds above $15\,\mathrm{m\,s^{-1}}$. Li et al. (2019) compared five bias correction methods for regional climate model (RCM) wind output over East Asia, finding that QM based on a fitted Weibull distribution generally outperforms linear scaling, variance scaling, and empirical QM in reducing higher-order statistical biases (skewness, kurtosis, wind power density).

A recognised drawback of standard QM is that it does not preserve the climate change signal when applied to future projections: it can artificially inflate or deflate projected trends, particularly at extreme quantiles (Maraun 2016; Maraun et al. 2017). This has motivated several variants. Equidistant QM (Li, Sheffield and Wood 2010) adjusts the correction function to account for projected distributional shifts. Quantile delta mapping (QDM) (Cannon, Sobie and Murdock 2015) preserves relative changes in each quantile between historical and projection periods, and is preferred when downstream health or hazard applications weight tail behaviour (Qian and Chang 2021). Scaled distribution mapping (SDM) (Switanek et al. 2017) takes a conceptually similar approach while also accounting for the frequency of non-zero events. The ISI-MIP approach (Hempel et al. 2013) and ISIMIP3BASD (Lange 2019) are implementations that preserve absolute changes in monthly temperature and relative changes in precipitation and other variables.

Empirical comparisons of correction methods do not consistently favour a single approach. Beyer, Krapp and Manica (2020) evaluated the delta method, generalised additive models (GAMs), and QM against a large global dataset of palaeoclimate reconstructions and found that the delta method performs marginally better for minimising median absolute bias, though spatial and temporal variation in performance is large. Teutschbein and Seibert (2012) reviewed methods for hydrological climate-change impact studies and found that while all methods correct the mean, clear differences exist in their capacity to correct standard deviation and extreme percentiles.

2.3.3 Distributional approaches for extremes.

Standard bias correction methods designed around the mean or moderate quantiles may misrepresent distributional tails. Sanabria et al. (2022) proposed a method that explicitly corrects both body and tail of the simulated distribution, applied to fire weather indices in Australia. Turkman et al. (2021) proposed a conditional quantile matching approach that calibrates the bulk and extremes of spatial wind speed data simultaneously, drawing on extreme value theory without imposing a fixed threshold. Lee et al. (2025) developed a parameter-space calibration method for extremes based on a linear mapping of parameters of a mixture model (Weibull body, generalised Pareto tail), applied to river runoff projections. Byun and Hamlet (2024) introduced an extrapolation procedure for empirical QM that handles future extreme values outside the historical CDF range while preserving the rank order of simulated extremes.

2.3.4 Multivariate methods.

All of the above approaches operate on a single variable at a time, leaving inter-variable and spatial dependence uncorrected. Vrac and Friederichs (2015) introduced empirical copula bias correction (EC-BC), which combines univariate QM with a shuffling technique to restore an empirical multivariate copula, improving inter-variable, spatial, and temporal dependence simultaneously. Cannon (2017) generalised QM to an N-dimensional probability density function transform (MBCn), demonstrating substantial improvements over univariate QM and pairwise correlation methods for fire weather indices. Mehrotra and Sharma (2016) developed a multivariate recursive quantile nesting bias correction (MRQNBC) to address auto- and cross-dependence across multiple time scales. These methods are computationally heavier than univariate corrections but are necessary when multivariate quantities such as fire weather or compound energy droughts are the target.

2.3.5 Statistical downscaling.

Bias correction and statistical downscaling are related but distinct tasks. Downscaling also increases spatial resolution by learning a mapping from coarse to fine scale. Classical methods include bias correction and spatial disaggregation (BCSD) (Maurer and Hidalgo 2008) and quantile mapping applied to RCM precipitation (Themeßl, Gobiet and Leuprecht 2011; Gudmundsson et al. 2012; Piani, Haerter and Coppola 2009; Piani et al. 2010). More recent deep learning approaches include stacked super-resolution convolutional networks (DeepSD) (Vandal et al. 2017) and generative adversarial networks (Leinonen, Nerini and Berne 2020). Machine learning downscaling of ERA5 wind speed with topographic features (Hu et al. 2023) and random forest calibration of ERA5 at upper model levels in complex terrain (Contreras et al. 2026) are among the most recent applied contributions. A comparison of machine learning statistical downscaling against six RCMs over Europe found that the statistical model had better scores for most standard metrics before bias correction, and comparable scores after it (Hernanz et al. 2023).

2.3.6 Bayesian and spatial-statistical approaches.

A smaller but growing strand of literature applies spatially explicit probabilistic frameworks. Iizumi et al. (2012) used Bayesian downscaling with Markov chain Monte Carlo (MCMC) to estimate global solar radiation in data-sparse regions. Ingebrigtsen, Lindgren and Steinsland (2014) and Ingebrigtsen et al. (2015) demonstrated non-stationary spatial models based on the stochastic partial differential equation (SPDE) approach with integrated nested Laplace approximation (INLA) (Rue, Martino and Chopin 2009a; Lindgren, Rue and Lindström 2011a) for annual precipitation. The SPDE-INLA combination offers $O(n^{3/2})$ computational complexity compared with $O(n^3)$ for dense Gaussian processes (Krainski et al. 2018). Villejo et al. (2025) applied this framework to data fusion of Philippine meteorological observations with numerical weather forecasts, modelling a spatially structured additive bias alongside a multiplicative bias. Carter, Chacón-Montalván and Leeson (2025) proposed a Bayesian hierarchical bias correction model that assumes shared latent Gaussian processes between in-situ observations and model output, with the aim of partially preserving the spatial covariance structure of the model after correction. Fioravanti et al. (2023) applied the INLA-SPDE approach to produce gridded temperature climatologies over Italy with propagated uncertainty. In the context of wind speed, Organ et al. (2025) incorporated bias-corrected personal weather station data into a Bayesian hierarchical spatiotemporal model, achieving a 7% reduction in prediction error relative to station-only models.

2.3.7 Forecast post-processing.

A related body of work addresses statistical post-processing of ensemble weather forecasts rather than reanalysis calibration. Distributional regression frameworks, including ensemble model output statistics (EMOS) and Bayesian model averaging (BMA), correct the distribution of forecast errors and improve calibration in the sense of Gneiting, Balabdaoui and Raftery (2007). Rasp and Lerch (2018) showed that neural networks for post-processing 2-m temperature significantly outperform benchmark EMOS methods. Barnes (2022) developed a multivariate Bayesian post-processing framework using directed acyclic graphs, applied to winter surface temperatures over the UK.

2.4 Validation metrics and evaluation

The literature employs a range of metrics to evaluate calibration and bias correction. Deterministic point metrics are most common: mean bias error (MBE), RMSE, mean absolute error (MAE), and the Pearson correlation coefficient r. Gneiting (2011) establishes the theoretical basis for the choice of scoring function, noting that the choice must match the target functional (mean, quantile, etc.).

and that scoring functions can be inconsistent for functionals other than the one they implicitly target. For probabilistic forecasts, the continuous ranked probability score (CRPS), reliability diagrams, and the probability integral transform (PIT) histogram are standard (Gneiting, Balabdaoui and Raftery 2007; Gneiting et al. 2008). Gneiting and Resin (2022) generalised these concepts to a universal coefficient of determination R^* applicable to any regression functional. Skill scores normalise raw scores against a reference (e.g. climatology or a simpler model) and are used in (Torralba et al. 2017) to evaluate seasonal wind speed predictions.

2.5 Reanalysis bias correction for wind energy: key products and benchmarks

The Global Wind Atlas (GWA) provides mean wind speed estimates at high resolution derived from the WAsP microscale model driven by ERA5 or MERRA-2. Gruber et al. (2019) assessed GWA-based mean bias correction of MERRA-2 for Brazilian wind power at four spatial levels, finding that GWA improves results at state and national level but not at the wind park level. Gruber et al. (2022) extended this assessment to multiple countries (US, Brazil, South Africa, New Zealand) and found that ERA5 outperforms MERRA-2 on error metrics, but that GWA3-based bias correction is actually detrimental to simulation quality, a finding that warrants attention for practitioners. Murcia et al. (2022) validated European-scale wind generation simulations from ERA5, the New European Wind Atlas (NEWA), and DTU's EIWR against 32 tall-mast wind speed observations and four years of country-level power production data. They found that NEWA-GWA2 and ERA5-GWA2 achieve the highest accuracy and precision for simulating national-level wind generation.

The New European Wind Atlas (NEWA) was produced by downscaling ERA5 with the Weather Research and Forecasting (WRF) mesoscale model at 3 km resolution, followed by further microscale downscaling with WAsP (Dörenkämper et al. 2020). WRF downscaling reduced the mean wind speed bias from $-1.50 \pm 1.30\,\mathrm{m\,s^{-1}}$ for ERA5 to $0.02 \pm 0.78\,\mathrm{m\,s^{-1}}$. The Wind Integration National Dataset (WIND) Toolkit provides a comparable product for the US (Draxl et al. 2015).

2.6 Observed trends and climate context

Reanalysis data have been used to characterise long-term wind speed trends. A global analysis found that terrestrial wind stilling , the observed decline in near-surface wind speeds over land since the 1980s , reversed around 2010, with implications for wind energy production projections (Zeng et al. 2019). Bett, Thornton and Clark (2017) used 142 years of the Twentieth Century Reanalysis (20CR) calibrated to ERA-Interim to show that many regions of interest for wind energy exhibit large multidecadal variability but very weak trends, suggesting that decade-scale climate variability is an important uncertainty in wind resource assessment. A review of EURO-CORDEX ensemble projections found projected increases in wind energy potential over northern Europe and decreases over southern Europe under future climate scenarios (Moemken et al. 2018).

3 Application to Renewable Energy Estimation and Derived Meteorological Quantities

3.1 Wind power simulation

Wind power output is derived from wind speed observations or model output via a turbine power curve, which maps hub-height wind speed to electrical power. The mapping is approximately cubic between cut-in and rated wind speed, and is zero outside the operational range. Because the power curve is nonlinear, biases in the wind speed distribution translate nonlinearly to biases in energy

output; a mean wind speed bias of +10% in a regime near rated speed implies a much smaller percentage bias in power than the same bias at lower wind speeds. This motivates correcting the full distribution rather than just the mean.

Staffell and Pfenninger (2016) reported the first international validation of MERRA/MERRA-2 for wind energy across 23 European countries. National correction factors were sufficient to bring hourly correlations above $R^2 = 0.95$, but the underlying spatial biases were large: overestimation of up to 50% in northwest Europe and underestimation of 30% in the Mediterranean. Olauson (2018) demonstrated that ERA5 outperforms MERRA-2 for both country-level and turbine-level wind generation modelling in Sweden, with approximately 20% lower MAE and RMSE, and 20% lower uncertainty in long-term energy yield correction. Benmoufok et al. (2024) showed that the spatial and temporal resolution of the correction factors matters: spatially resolved cluster-based correction of ERA5 across Danish turbines reduced monthly capacity factor error by 43% relative to uncorrected ERA5. Wang et al. (2026) applied a cluster-based correction across 319 UK wind farms, reducing monthly simulation errors by more than 32%, and provided maps of the spatial structure of ERA5 bias over the UK, identifying the Scottish Highlands and Welsh mountains as regions of systematic underestimation.

Long-term (greater than 40 year) hourly time series of offshore wind generation for European wind farms were constructed by Hayes, Stocks and Blakers (2021) using ERA5, reducing hourly and daily capacity factor RMSE by 10% and 18%, respectively, compared with the Renewables.ninja platform. For the continental US, Wilczak et al. (2024) found that ERA5-derived wind power is biased low by approximately 20% on an annual basis, and proposed linear regression corrections for wind speed alongside quantile corrections for solar power. For Brazil, Aquino Ferreira, Cyrino Oliveira and Maçaira (2022) evaluated MERRA-2 with interpolation, extrapolation, and bias correction strategies, confirming that treatment of the dataset is necessary before use for wind resource assessment.

Limitations of reanalysis for wind power include the poor representation of the stable nocturnal boundary layer (Davidson and Millstein 2022; Holtslag et al. 2013), the mismatch between reanalysis grid spacing and topographic variability at wind farm sites (Gualtieri 2022; Contreras et al. 2026), and the fact that standard vertical extrapolation based on Monin-Obukhov similarity theory (a logarithmic wind profile corrected for atmospheric stability) becomes unreliable above the surface layer under stable stratification (Optis, Monahan and Bosveld 2016). Machine learning approaches to vertical extrapolation and downscaling, including random forests (Contreras et al. 2026) and topographic regression (Hu et al. 2023), are increasingly used to address the resolution limitation.

3.2 Solar PV estimation

Global horizontal irradiance (GHI, measured in $\mathrm{W\,m^{-2}}$) is the primary input to PV power simulations. Pfenninger and Staffell (2016) combined MERRA/MERRA-2 and the Meteosat-based CM-SAF SARAH satellite dataset to produce 30-year hourly PV simulations for Europe, validated against output from more than 1000 PV systems and national aggregates. Satellite data showed slightly better accuracy; reanalysis data showed greater temporal stability. A simple mean bias correction was applied by matching the simulated to the observed mean bias at each location.

Wilczak et al. (2024) evaluated ERA5-derived solar power over the contiguous US and found it biased high by an average of 23% in winter and by less on an annual basis. The bias was especially large for multi-day cloudy periods, where ERA5 solar power exceeded observations by 62% at 10-day duration during the most extreme negative anomaly events. A quantile correction substantially reduced these errors.

Urraca et al. (2018) evaluated ERA5 and the regional reanalysis COSMO-REA6 for GHI

estimation against 41 BSRN stations globally and 294 stations in Europe. ERA5 outperforms ERA-Interim and MERRA-2 across most metrics, but both products degrade at coastal sites and mountains. ERA5 overestimates under cloudy conditions and slightly underestimates under clear skies, pointing to cloud fraction prediction as the primary remaining error source. Bailey et al. (2024) proposed a physically constrained modification of QM for global horizontal irradiance that incorporates clear-sky GHI to ensure physically plausible corrected values.

Bett and Thornton (2016) used reanalysis data to characterise the joint distribution of wind and solar supply in Britain, finding a weak anticorrelation ($-0.4 \lesssim \rho \lesssim -0.2$) that implies partial but insufficient natural balancing. The scenario with least seasonal variability was approximately 70% solar to 30% wind capacity, a result directly relevant to grid integration planning.

3.3 Wind and solar assessment for power system planning

Several studies have assembled long-term reanalysis-based reconstructions of demand and renewable supply at the national or continental scale. These reconstructions underpin power system planning studies that require multi-decadal time series to capture inter-annual variability. Bloomfield et al. (2016) showed that fewer than 10 years of supply and demand data are insufficient to characterise the climate-driven variability of the GB power system, and that a 2025 wind scenario would amplify this variability fourfold relative to a purely thermal system. Bloomfield et al. (2018) and Bloomfield et al. (2020) traced the changing meteorological sensitivity of the GB power system from a temperature-dominated to a wind-dominated regime as installed wind capacity grew. Cannon et al. (2014) used 33 years of MERRA data to quantify extreme wind power generation events in Great Britain, finding that the frequency of prolonged low-wind events follows a Poisson-like process. Potisomporn, Adcock and Vogel (2023) evaluated ERA5 near-surface wind speed against 205 UK stations, finding systematic biases in wind speed standard deviation (approximately -0.35 to $-0.43\,\mathrm{m\,s^{-1}}$) that lead to underestimation of short-duration low-wind events.

Seasonal forecasting of renewable supply is an adjacent application in which calibration of forecast bias is also essential. Torralba et al. (2017) compared simple bias correction and calibration (mean-scaling of the forecast distribution to match observed climatology and variance) for ECMWF seasonal wind speed predictions, finding that the calibration step is indispensable for producing statistically consistent predictions. Clark et al. (2017) demonstrated that the predictability of the North Atlantic Oscillation (NAO) one season ahead translates to skilful winter wind speed and demand forecasts for Europe.

3.4 Wave energy and other derived quantities

Reanalysis-driven wave models provide estimates of significant wave height (SWH), energy period, and wave power density for marine renewable energy assessment. Pérez-Molina and Carta (2025) applied several machine learning methods to extend wave energy estimates at a Canary Islands buoy from a short observational record using ERA5 data, finding that multiple linear regression and support vector regression achieved MAE of $2.56\,\mathrm{kW\,m^{-1}}$ compared with $4.38\,\mathrm{kW\,m^{-1}}$ for direct ERA5 use. The superiority of even simple regression calibration over raw reanalysis in wave applications echoes findings from wind energy.

Reanalysis data are also used to estimate hydrological quantities. Berg et al. (2003) showed that bias-corrected ECMWF and NCEP-NCAR reanalysis products forced a land surface model to produce improved soil moisture, runoff, and snow water equivalence simulations over North America. Fuka et al. (2014) demonstrated that CFSR precipitation and temperature can produce stream discharge simulations comparable to or better than those forced with traditional rain gauge networks, particularly when gauges are more than 10 km from the watershed. The WFDE5 dataset (Cucchi et al. 2020) provides bias-adjusted ERA5 surface variables at 0.5° resolution for impact

studies, using the WATCH Forcing Data (WFD) methodology to adjust precipitation totals against gridded observational products; this results in more plausible global hydrological water balance components in an uncalibrated hydrological model.

Wind speed reconstructions from reanalysis have been applied to estimate wind power density (WPD) globally. Hou et al. (2025) identified that using daily rather than hourly wind speed data leads to an average underestimation of WPD by 35.6% globally, and proposed a calibration coefficient to harmonise WPD estimates across temporal resolutions.

3.5 Identified gaps

Several gaps remain prominent in the literature. Uncertainty quantification for the horizontal interpolation step,-the translation from a regular reanalysis grid to the precise latitude and longitude of a generating site,-is typically addressed by bilinear interpolation without propagated error bounds. The vertical interpolation from the reanalysis reference level (usually 10 m or 100 m) to turbine hub height introduces further uncertainty through the wind shear profile, which depends on atmospheric stability and is frequently approximated by a deterministic power law. Few studies quantify the joint uncertainty from both interpolation steps simultaneously. At the power conversion step, deterministic power curves (the function relating wind speed to power output for a specific turbine model) are widely used despite evidence that they mask substantial variability due to turbulence, air density, and yaw error Yan et al. (2019) and Xu et al. (2021). The literature on distributional calibration of power output, as opposed to calibration of the input wind speed, remains sparse relative to the attention given to point forecasting.

4 Data

This study draws on four data sources: half-hourly metered generation from the Balancing Mechanism Reporting Service (BMRS) operated by Elexon, wind farm metadata from the Renewable Energy Planning Database (REPD), hourly wind speed fields from the ERA5 global reanalysis, and generic power curves derived from the Global Wind Atlas (GWA). The following subsections describe each source, the spatial and temporal alignment procedure, the vertical extrapolation of reanalysis wind speed to turbine hub height, and the conversion of hub-height wind speed to capacity factor. Table 1 summarises the key attributes of each source.

4.1 Wind Power Generation Data (Elexon BMRS)

Generation data are obtained from the Balancing Mechanism Reporting Service (BMRS), which is maintained by Elexon as part of the Balancing and Settlement Code (BSC) for the Great Britain (GB) electricity market Elexon 2024. The BMRS collects and publishes operational metering data for generators connected to the transmission network. The unit of observation is the Balancing Mechanism Unit (BMU): a distinct registered entity within the settlement system, which in the case of wind generation typically corresponds to a single wind farm or a cluster of wind farms operating under a common metering arrangement. Generation is reported at half-hourly resolution, aligned with the 30-minute settlement period used in GB electricity markets, in units of megawatts (MW) averaged over each period.

The dataset covers the period from January 2019 to December 2025. Only BMUs classified as wind generation are retained. It is important to note that the BMRS captures generation from plants above the transmission connection threshold; smaller embedded generators, which do not export directly to the transmission network, are not represented. The generation figures therefore correspond to metered output rather than total installed capacity.

4.2 Wind Farm Metadata (REPD)

Geographic and technical metadata for wind farms are obtained from the Renewable Energy Planning Database (REPD), published on a quarterly basis by the Department for Energy Security and Net Zero (DESNZ) Department for Energy Security and Net Zero 2025. The REPD tracks renewable electricity projects from planning application through to operational status across the UK and records, for each project, its location (latitude and longitude of the site centroid), technology type, installed capacity in megawatts, turbine count, turbine model, and, where available, hub height in metres. Barbour ABI manages the database on behalf of DESNZ and the data are released under the Open Government Licence.

Each Elexon BMU is matched to one or more REPD entries by site name and location, following the approach described in Staffell and Pfenninger 2016. The match associates each BMU with a set of geographic coordinates and the corresponding installed capacity, which is required for computing capacity factors (Section 4.5). Hub height is available for approximately a third of the matched wind farms. For the remaining farms, hub height is imputed as the mean hub height of all matched farms of the same type, computed separately for onshore and offshore sites. This imputation introduces uncertainty into the vertical extrapolation described in Section 4.4, since hub heights vary across turbine generations and manufacturers.

4.3 ERA5 Reanalysis Wind Speed

Wind speed data are taken from the ERA5 global reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) Hersbach et al. 2020. A reanalysis is the output of a data assimilation system that ingests historical observations into a fixed numerical weather prediction model to produce a spatially complete, physically consistent record of past atmospheric states. ERA5 provides hourly estimates of a large number of atmospheric variables at approximately 31 km horizontal resolution on a regular $0.25^\circ \times 0.25^\circ$ latitude-longitude grid.

For this study, hourly 10-m and 100-m wind speed fields are extracted for the UK domain, spanning January 2019 to December 2025. Wind speed at each grid point is computed from the zonal (u) and meridional (v) wind components as $|\mathbf{u}| = (u^2 + v^2)^{1/2}$.

4.3.1 Spatial matching.

For each wind farm in the matched REPD-Elexon dataset, the ERA5 grid point whose latitude-longitude coordinates minimise the great-circle distance to the farm centroid is identified, and its hourly wind speed time series is extracted. Using the nearest grid point rather than spatial interpolation is standard practice in the reanalysis wind energy literature Staffell and Pfenninger 2016; Gualtieri 2022: it avoids introducing artificial correlations between adjacent grid points that would result from linear interpolation weighting, and is consistent with the approximately 28 km grid spacing at UK latitudes, which is already coarser than most individual wind farms. The nearest grid point is held fixed for the full analysis period.

4.4 Vertical Extrapolation to Hub Height

ERA5 provides wind speed at reference heights of 10 m (above surface level) and 100 m (above surface level). Modern onshore turbines typically operate at hub heights of 80,140 m, and offshore turbines at 80,120 m. Extrapolation from the available reference level to hub height is therefore required. The power law wind profile is used for this purpose. Let $U(z)$ denote the mean wind speed at height z above the surface. Given a reference wind speed $U(z_{\mathrm{ref}})$ at reference height z_{ref}, the power law gives

$$U(z) = U(z_{\text{ref}}) \left(\frac{z}{z_{\text{ref}}} \right)^{\alpha}, \tag{1}$$

where α is a dimensionless wind shear exponent that governs the rate of increase of wind speed with height. The power law and the logarithmic wind profile are the two most widely used models for vertical extrapolation of wind speed from a reference height to turbine hub height in wind energy applications Gualtieri 2022; Jung and Schindler 2021.

The value of α depends on atmospheric stability, surface roughness, and terrain complexity. Under neutral stability over open, flat terrain, α is approximately $1/7 \approx 0.143$ for land-based sites and approximately 0.10,0.11 for offshore sites Jung and Schindler 2021. Applying a constant α is a simplification: in practice, α varies diurnally and seasonally, and deviations from the constant value can introduce errors of several per cent in hub-height wind speed estimates, with correspondingly larger errors in estimated wind power density due to the cubic speed-power relationship Jung and Schindler 2021. Despite this, a constant exponent is commonly assumed in large-scale reanalysis-based wind energy studies because long-term, multi-height wind speed measurements required to estimate site-specific α values are rarely available across an entire wind fleet.

In this study, $\alpha = 0.14$ is applied for onshore wind farms, and $\alpha = 0.11$ for offshore wind farms, consistent with the empirical values reviewed in Jung and Schindler 2021. Where ERA5 100-m wind speed is available, $z_{\text{ref}} = 100\,\text{m}$ is used as the reference level, reducing the extrapolation distance and thus the sensitivity to the choice of α; 10-m wind speed is used as a fallback. The hub height z is taken from the REPD record, where available and otherwise from the imputed values described in Section 4.2. The limitation introduced by hub height imputation propagates directly into the extrapolated wind speed: since Equation 1 is nonlinear in z, farms with substantially different true hub heights from the imputed value will have systematically biased extrapolated wind speeds.

4.5 Wind Speed to Capacity Factor Conversion

Hub-height wind speeds are converted to normalised power output using generic power curves derived from those used in the Global Wind Atlas (GWA) Technical University of Denmark and World Bank Group 2023; Dörenkämper et al. 2020. A power curve specifies the relationship between hub-height wind speed and the electrical power output of a turbine, expressed as a fraction of its rated (nameplate) capacity. This fraction is referred to as the capacity factor (CF) and takes values in $[0, 1]$. Power curves are nonlinear: output is zero below the cut-in wind speed (typically around $3\,\text{m}\,\text{s}^{-1}$), increases approximately as the cube of wind speed through the operating range, reaches rated output at the rated wind speed (typically $12{,}14\,\text{m}\,\text{s}^{-1}$), and returns to zero above the cut-out wind speed (typically around $25\,\text{m}\,\text{s}^{-1}$) for safety reasons.

A single generic onshore power curve and a single generic offshore power curve are applied, irrespective of the specific turbine model installed at each farm. This is a standard simplification in fleet-scale analyses Staffell and Pfenninger 2016: individual turbine specifications are frequently unavailable, and the use of a representative curve avoids the need for proprietary manufacturer data. The resulting capacity factor time series are compared with observed capacity factors computed from the Elexon BMRS generation data divided by the REPD installed capacity. The use of generic curves introduces uncertainty for farms whose turbines have performance characteristics that deviate materially from the representative curve, in particular for older or smaller onshore turbines and for the most recent large-scale offshore turbines.

Table 1: Summary of data sources used in this study.

Source	Variable(s)	Temporal coverage	Coverage
Elexon BMRS	Half-hourly generation (MW) per BMU	2019–2025	Great Britain
REPD	Location, capacity, turbine count, hub height	Updated quarterly	United Kingdom
ERA5	Hourly wind speed at 10 m and 100 m	2019–2025	Global
Global Wind Atlas	Generic onshore and offshore power curves	Long-term average	Global

5 Models

5.1 Calibration of the reanalysis wind speed estimate

Before site-level power curve modelling is applied, the ERA5-derived power estimate is calibrated against metered generation at the level of Great Britain (GB) as a whole. The aggregate capacity factor at time t is computed as

$$p_{\mathrm{tot},t} = \frac{\sum_i p_{it}}{\sum_i C_i}, \tag{2}$$

where p_{it} is the power output of farm i at time t and C_i is its installed capacity. This normalisation produces an aggregate capacity factor in $[0, 1]$ that is comparable across periods of different fleet composition.

Four calibration models of increasing complexity are considered.

5.1.1 Linear model.

The simplest calibration regresses the observed aggregate capacity factor on the ERA5-derived power estimate $\hat{p}_t$, with main effects selected by a stepwise Akaike information criterion (AIC) procedure. The model includes an intercept varying by turbine type k (onshore or offshore), month m, and hour h; a slope varying by type and month; and a cubic wind speed correction term. The calibrated power estimate is

$$p_t^{\mathrm{calib}} = \alpha_{k,m,h} + \beta_{k,m}\,\hat{p}_t + \sum_{j=1}^{3} \phi_j\, w_t^j, \tag{3}$$

where $\alpha_{k,m,h}$ is the intercept for type k, month m, and hour h; $\beta_{k,m}$ is a multiplicative correction factor for type and month; w_t is the ERA5 wind speed at time t; and ϕ_j are coefficients on a cubic polynomial in wind speed. The polynomial terms in w_t allow the correction to vary with wind speed regime, capturing systematic mis-specification in the power curve shape under the generic curve approximation.

5.1.2 Autoregressive model with random effects.

The linear model assumes independent residuals. To account for temporal persistence in the calibration error, an autoregressive term of order one (AR(1)) is added, and the intercept and slope terms are replaced by structured random effects. Inference is carried out using the Integrated Nested Laplace Approximation (INLA) Rue, Martino and Chopin 2009b, implemented in the R-INLA package Lindgren, Rue and Lindström 2011b. The model is

$$p_t^{\text{calib}} = \alpha_0 + \sum_{e \in \{k,m,h\}} \alpha_e + \beta_k \, \hat{p}_t + f_w(w_t) + u_t, \tag{4}$$

where α_0 is a global intercept; the α_e are random effects for type (k), month (m, modelled as cyclic), and hour of day (h, modelled as cyclic); β_k is a random slope for type; f_w is a smooth effect of wind speed modelled as a second-order random walk (RW2) over discretised wind speed bins; and $u_t \sim \text{AR}(1)$ captures residual temporal autocorrelation. The cyclic structure on month and hour ensures continuity at the boundary of the respective periods. The AR(1) component u_t satisfies $u_t = \rho \, u_{t-1} + \varepsilon_t$ with $|\rho| < 1$ and $\varepsilon_t \sim \mathcal{N}(0, \sigma_u^2)$.

5.1.3 Hierarchical Bayesian smooth model.

A more flexible formulation replaces the multiplicative correction with smooth functions of the ERA5 power estimate and wind speed:

$$p_t^{\text{calib}} = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_t) + s_w(w_t), \tag{5}$$

where α is a global intercept; β_k, β_m, and β_h are iid Gaussian random effects for type, month, and hour, respectively; s_p and s_w are smooth functions modelled as RW2 components over their respective domains. This formulation allows the correction to be non-monotone in both the power estimate and wind speed, without restricting the functional form to a polynomial. Inference is again performed via INLA.

5.1.4 Quantile mapping.

Quantile mapping is included as a benchmark. Rather than modelling the conditional mean, this approach transforms the ERA5-derived power estimate so that its marginal distribution matches that of the observed generation. Formally, the calibrated estimate is

$$p_t^{\text{calib}} = F_{\text{obs}}\big(F_{\text{mod}}^{-1}(p_t)\big) \, , \tag{6}$$

where F_{mod} is the empirical cumulative distribution function (CDF) of the ERA5-derived power estimate and F_{obs} is the empirical CDF of the observed generation. The composite map $F_{\text{obs}} \circ F_{\text{mod}}^{-1}$ acts as a monotone transfer function that corrects distributional bias without imposing any parametric assumption on the correction shape.

5.2 Power curve model

The site-level power curve model is fitted to individual wind farms. A wind turbine power curve is the function mapping hub-height wind speed to electrical output, expressed as a fraction of rated capacity. Because observed output at any given wind speed is a mixture of zero (turbine offline or below cut-in speed) and positive values (turbine generating), a two-component model is required.

The model is a Zero-Inflated Beta (ZIB) regression with a penalisation component that anchors the estimated curve towards the generic power curve derived from the Global Wind Atlas. R-INLA is used for inference Rue, Martino and Chopin 2009b; Lindgren, Rue and Lindström 2011b. The three components are as follows.

Zero-probability component

The probability that observed output is zero at time t for farm i, denoted π_{it}, is modelled on the logistic scale:

$$\eta_{it}^{(0)} = \alpha_{\text{zero}} + f_{\text{zero}}(w_{it},\, i), \tag{7}$$

$$Z_{it} \mid \eta_{it}^{(0)} \sim \text{Bernoulli}(\pi_{it}), \qquad \pi_{it} = \frac{1}{1+\exp\{-\eta_{it}^{(0)}\}}, \tag{8}$$

where α_{zero} is a global intercept for the zero component and $f_{\text{zero}}(w_{it}, i)$ is a smooth function of wind speed w_{it} at farm i. This component captures the fact that zero output occurs both at very low wind speeds (below cut-in) and at very high wind speeds (above cut-out), producing a non-monotone probability of zero generation.

Positive-output component

Conditional on positive output, the capacity factor is modelled with a Beta distribution:

$$\eta_{it}^{(\beta)} = \alpha_{\text{pc}} + f_{\text{pc}}(w_{it},\, i), \tag{9}$$

$$P_{it}^{\text{obs}} \mid \eta_{it}^{(\beta)} \sim \text{Beta}(\mu_{it}^{+},\, \phi), \qquad \mu_{it}^{+} = \frac{1}{1+\exp\{-\eta_{it}^{(\beta)}\}}, \tag{10}$$

where α_{pc} is a global intercept for the positive component, $f_{\text{pc}}(w_{it}, i)$ is a smooth function of wind speed at each farm, $\mu_{it}^{+} \in (0, 1)$ is the conditional mean capacity factor, and $\phi > 0$ is a precision parameter shared across farms.

Penalisation component

To regularise the smooth power curve $s^{\text{pc}}(w, i) \approx \mu_{it}^{+}$ towards the generic power curve $\widehat{PC}_i(w)$ available from the Global Wind Atlas, a pseudo-likelihood penalty is added. Denoting $d_{it} = |\widehat{PC}_i(w_{it}) - s^{\text{pc}}(w_{it}, i)|$, the penalty is equivalent to assuming

$$\widehat{PC}_i(w_{it}) \mid \eta_{it}^{(\beta)} \sim \mathcal{N}\left(\eta_{it}^{(\beta)},\, \tau_{\text{ps}}^{-1}\right), \tag{11}$$

where $\tau_{\text{ps}} > 0$ is a precision (penalisation) parameter. The full objective is

$$\max_{\theta}\; l^{(0)}(\theta) + l^{(\beta)}(\theta) - \tau_{\text{ps}} \sum_{i,t} d_{it}^2, \tag{12}$$

where $\theta = (\theta^0,\, \theta^{\text{pc}},\, \tau_{\text{ps}})$ collects the parameters of all three components, and $l^{(0)}(\theta)$ and $l^{(\beta)}(\theta)$ the likelihoods for zero and positive observations, respectively. Larger values of τ_{ps} pull the estimated curve closer to the generic reference; $\tau_{\text{ps}} = 0$ recovers the unpenalised ZIB model.

Smooth components via 1D SPDE

Both $f_{\text{zero}}(w, i)$ and $f_{\text{pc}}(w, i)$ are modelled using a one-dimensional Stochastic Partial Differential Equation (SPDE) representation over wind speed, with a uniform correlation structure across farms Lindgren, Rue and Lindström 2011b. The basis function expansion is

$$f(w, i) = \sum_{k} \beta_{ik}\, \psi_k(w), \tag{13}$$

where $\psi_k(w)$ are piecewise-linear basis functions defined over a mesh on the wind speed domain, and β_{ik} are coefficients for farm i. The marginal covariance between two wind speed values w_s and w_t at the same farm follows a Matérn family:

$$\mathrm{Cor}(f(w_s, i),\, f(w_t, i)) = \frac{1}{2^{\nu-1}\,\Gamma(\nu)}\,(\kappa d)^{\nu} K_{\nu}(\kappa d), \tag{14}$$

where $d = |w_s - w_t|$ is the distance in wind speed, $\kappa > 0$ is a range parameter, $\nu > 0$ is a smoothness parameter, and K_ν is the modified Bessel function of the second kind. The SPDE approach of Lindgren, Rue and Lindström 2011b provides a computationally efficient sparse matrix representation of this Gaussian field by linking it to a Gaussian Markov random field. Across farms $j \neq k$, a uniform correlation is assumed:

$$\mathrm{Cor}(f(w, j),\, f(w, k)) = \rho, \tag{15}$$

with $\rho \in [0, 1]$ estimated from data. This structure pools information across farms, which improves estimates for farms with limited data.

Expected power

The ZIB model yields the following point estimate of capacity factor at time t and farm i:

$$\hat{P}_{it} = \begin{cases} 0, & \text{with probability } \pi_{it}, \\ \mu^{+}_{it}, & \text{with probability } 1 - \pi_{it}, \end{cases} \tag{16}$$

with marginal expectation

$$\mathbb{E}[P_{it} \mid w_{it}, i] = (1 - \pi_{it}) \times \mu^{+}_{it}. \tag{17}$$

5.3 Modelling power curves with speed cut-out

5.3.1 Parametric power curve fitting

A more traditional alternative to the nonparametric ZIB model is a fully parametric representation of the power curve shape. The five-parameter logistic function introduced by Lydia et al. 2013 provides a flexible sigmoidal shape:

$$P(u) = D + \frac{A - D}{\left(1 + (u/C)^{B}\right)^{G}}, \tag{18}$$

where u is hub-height wind speed; A and D are the upper and lower asymptotes of the curve (corresponding to rated and zero power); C controls the wind speed at the inflection point (approximately the rated wind speed); B governs the slope at the inflection point; and G controls the asymmetry of the transition. For a physically meaningful curve, $A > D \geq 0$ and $B < 0$.

The five-parameter curve does not model cut-out behaviour: the curve approaches A monotonically as $u \to \infty$. To capture the decrease in output at high wind speeds above the cut-out threshold, two additional inflection parameters are introduced, giving a seven-parameter extension:

$$P(u) = D + \frac{A - D}{\left(1 + (u/C_1)^{B_1}\right)^{G}\left(1 + (u/C_2)^{B_2}\right)^{G}}, \tag{19}$$

where C_1 and B_1 govern the ascending part of the curve (as in the five-parameter model), C_2 is the wind speed at the cut-out inflexion point, $B_2 > 0$ controls the rate of decline after cut-out, and G applies to both transitions. Parameters are estimated by maximising the likelihood via `optim` in R, subject to the constraints $B_1 < 0$, $B_2 > 0$, and $A > D$, which are enforced by reparameterising on a log scale.

5.3.2 Pooling data for cut-in and cut-out speed estimation

The probability of zero generation at high wind speeds, which is informative about the cut-out threshold, is sparse at the level of individual wind farms: even over a five-year period (2019–2024), most farms record very few observations above $20\,\mathrm{m\,s^{-1}}$. To obtain stable estimates, data are pooled across farms using the same SPDE-based model structure as in Section 5.2, but restricted to wind speeds above a lower threshold so as to focus on the cut-off region.

The model for the probability of zero generation in the cut-off region takes the same zero-probability form as Equation (8):

$$\pi_{it}^{\mathrm{cut}} = \frac{1}{1+\exp\{-\eta_{it}^{(0,\mathrm{cut})}\}}, \qquad \eta_{it}^{(0,\mathrm{cut})} = \alpha_{\mathrm{cut}} + f_{\mathrm{cut}}(w_{it}, i), \tag{20}$$

where α_{cut} is a global intercept and $f_{\mathrm{cut}}(w, i)$ is a smooth function of wind speed shared across farms via the uniform correlation structure in Equation (15). The pooled estimate allows the SPDE smooth to borrow strength across farms with limited high-wind observations, producing a stable estimate of the wind speed at which the probability of zero generation begins to rise sharply.

6 Results

6.1 Calibration of the reanalysis wind speed estimate

The first tool to assess model fit is to look at the scatter plot of estimates versus actual power. Figure 1 shows a frequency plot to highlight where most of the data points lie, which usually in the low power regime, plotting the uncalibrated ERA5-derived power estimate against observed aggregate GB generation.

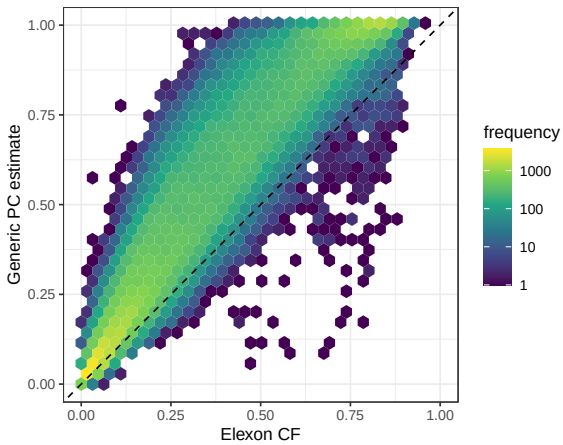

Figure 1: Uncalibrated ERA5-derived aggregate GB capacity factor versus observed generation before calibration. *(PENDING: add description.)*

Figure 4 shows results under the hierarchical Bayesian smooth model (Equation 5).

Figure 5 shows results under quantile mapping (Equation 6).

6.2 Power curve model

The ZIB-SPDE model was fitted to one year of data (2024) from a sample of 25 wind farms, excluding periods flagged as outages in the Regulatory Energy Market Information Transparency (REMIT) database. Figure 6 shows the expected capacity factor as a function of wind speed, integrating the zero-probability and positive-output components via Equation (17).

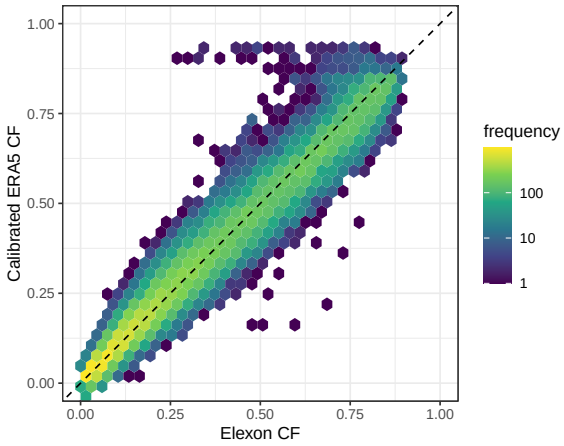

Figure 2: ERA5-derived aggregate GB capacity factor versus observed generation after linear calibration (Equation 3).

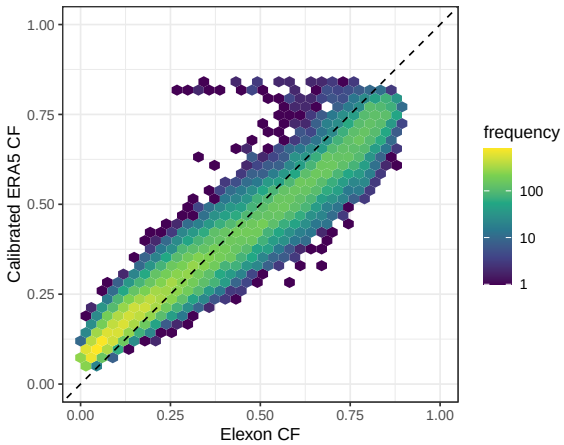

Figure 3: ERA5-derived aggregate GB capacity factor versus observed generation after AR(1) calibration (Equation 4).

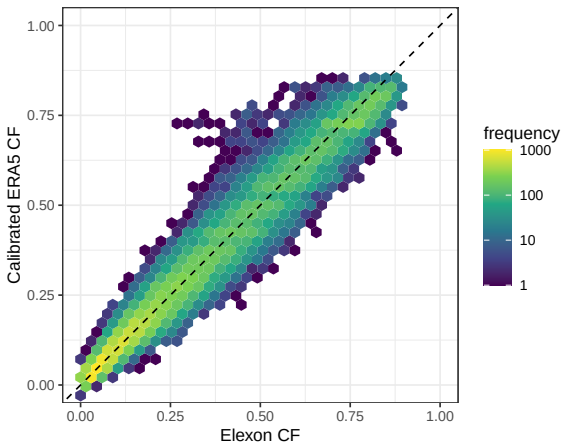

Figure 4: ERA5-derived aggregate GB capacity factor versus observed generation after hierarchical Bayesian calibration (Equation 5).

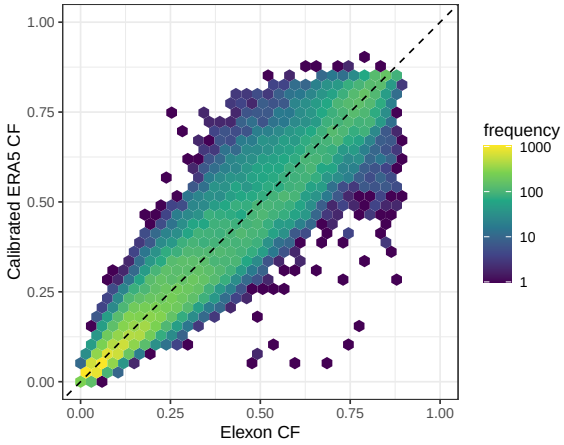

Figure 5: ERA5-derived aggregate GB capacity factor versus observed generation after quantile mapping calibration (Equation 6).

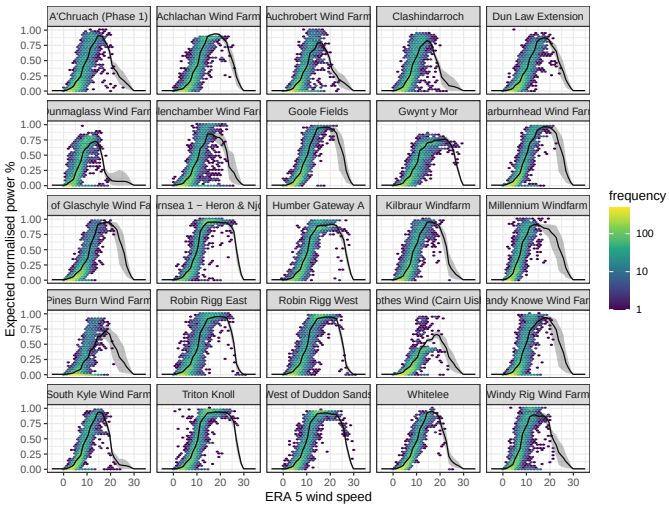

Figure 6: Expected capacity factor from the ZIB-SPDE model as a function of hub-height wind speed

Figures 7 and 8 show the estimated conditional mean power curve (μ_{it}^{+}) and the probability of zero generation (π_{it}) separately. The zero-probability component rises sharply at both low and high wind speeds, consistent with cut-in and cut-out behaviour.

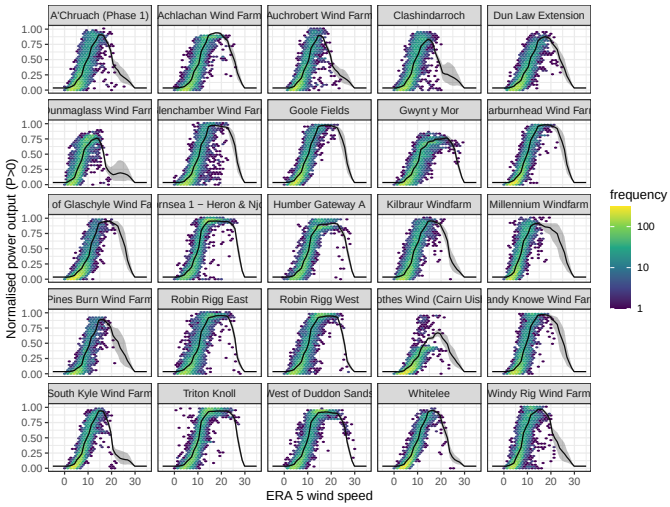

Figure 7: Estimated conditional mean power curve (μ^{+}) from the ZIB-SPDE model.

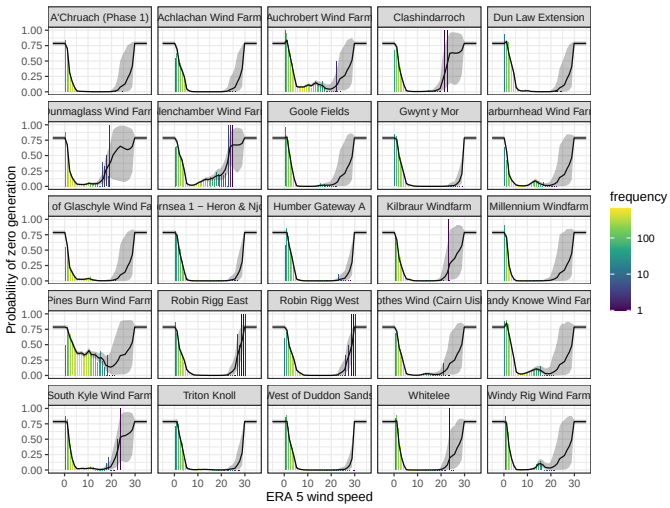

Figure 8: Estimated probability of zero generation (π) from the ZIB-SPDE model.

Figure 9 compares power curve estimates across the four model variants: a Beta model with RW2 smooth (B-RW2), a Beta model with 1D SPDE smooth (B-SPDE), a ZIB model with 1D SPDE smooth (ZIB-SPDE), and the penalised version (ZIB-SPDE-P). Zero-inflated models recover low-wind behaviour more accurately than Beta-only models, as the Beta distribution does not assign probability mass at exactly zero.

Figure 10 shows the aggregate calibration performance of each model variant against observed generation. Power curve modelling substantially reduces the scatter relative to the generic-curve baseline.

6.3 Parametric power curve fitting

The seven-parameter model (Equation 19) was fitted to a subsample of Scottish wind farms for which the 2024 data contain a sufficient number of observations at wind speeds above $20\,\mathrm{m\,s^{-1}}$ to identify the cut-out inflection. Figure 11 shows the fitted curves alongside the observed data.

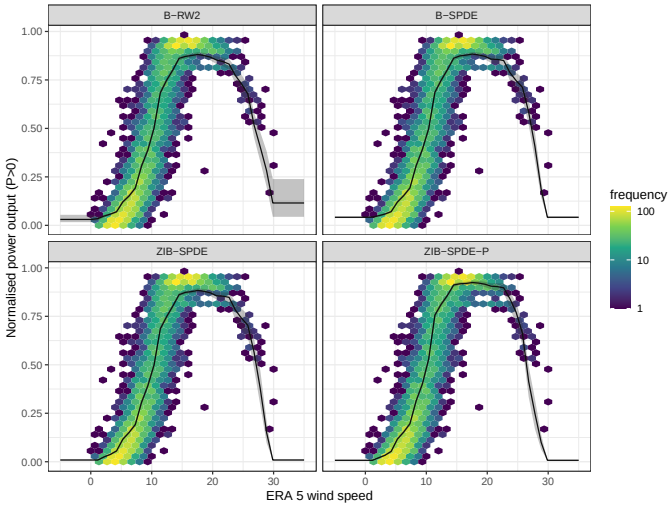

Figure 9: Power curve estimates under the B-RW2, B-SPDE, ZIB-SPDE, and ZIB-SPDE-P models.

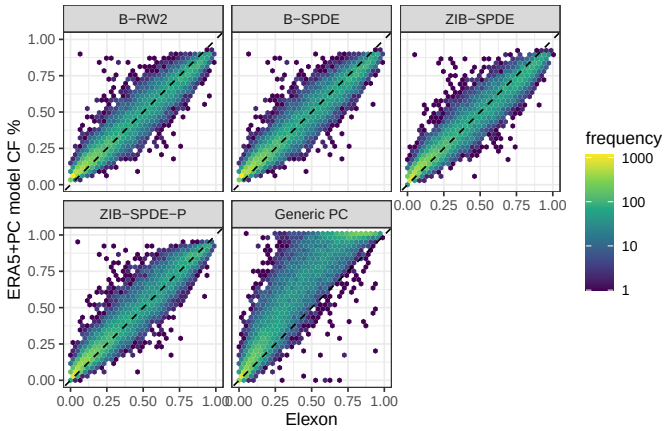

Figure 10: Predicted versus observed capacity factor for the power curve model variants.

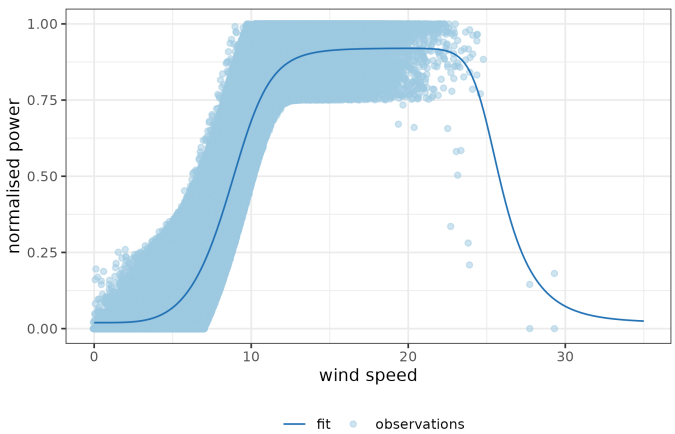

Figure 11: Seven-parameter power curve fits for Scottish wind farms.

6.4 Pooling data for cut-in and cut-out speed estimation

Figure 12 shows the estimated probability of zero generation as a function of wind speed in the cut-off region, pooled across farms.

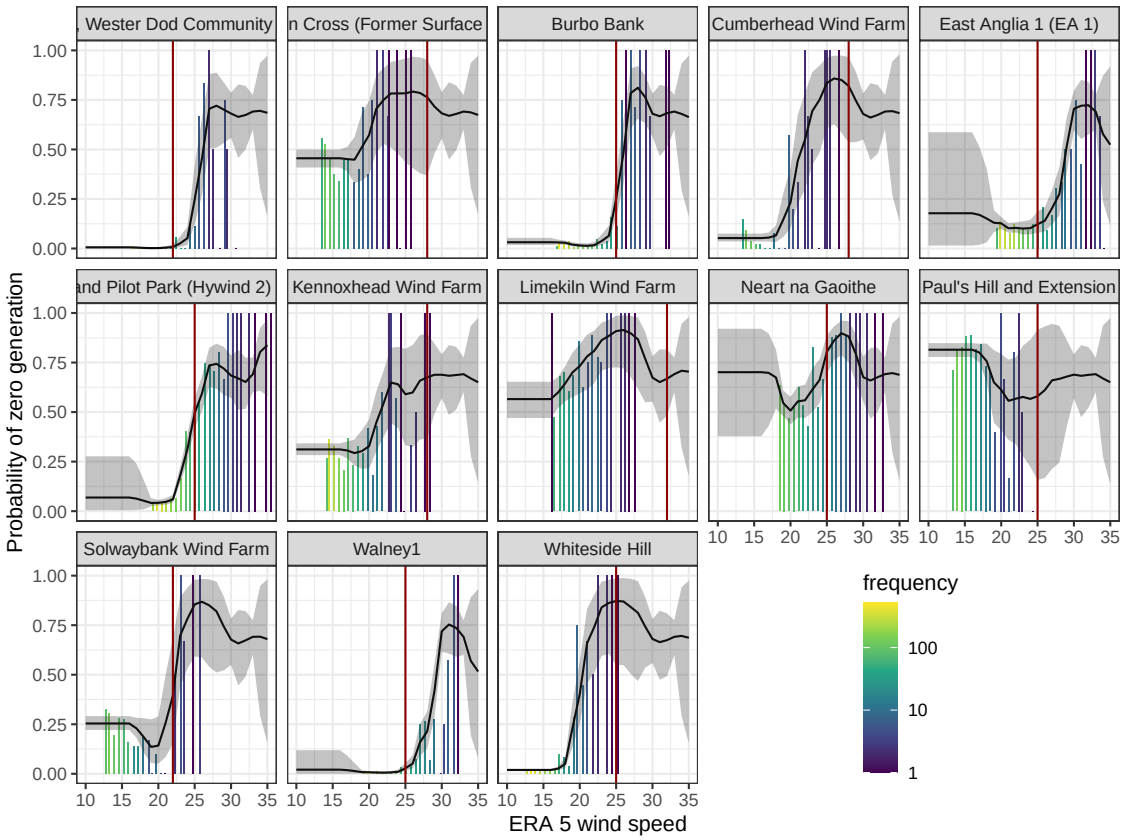

Figure 12: Pooled estimated probability of zero generation at high wind speeds from the cut-off SPDE model

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